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Psychiatr Serv. Author manuscript; available in PMC 2025 May 29.

Published in final edited form as:
Psychiatr Serv. 2023 March 01; 74(3): 312–315. doi:10.1176/appi.ps.202100608.

Racial Differences in Emergency Department Visit Characteristics and Management of Preadolescents at Risk of Suicide

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Abstract

Objective: Suicide rates and frequency of pediatric emergency department (ED) visits for suicidal thoughts and behaviors have increased among Black preadolescents in the United States in recent years. This study examined whether characteristics of ED visits and treatment management of preadolescents with suicidal thoughts and behaviors differed by race.

Methods: An electronic medical record query identified patients ages 8–12 (N=504) who visited a pediatric ED with a psychiatric-related chief complaint in 2019. The authors examined suicidal thoughts and behaviors that were reported with the Ask Suicide-Screening Questions tool, ED clinical impression, and ED disposition overall and by race.

Results: Compared with other racial groups, Black preadolescents were less likely to report suicidal thoughts, despite equivalent lifetime histories of suicide attempts, and were more likely to be brought to the ED by police and discharged (instead of being admitted to inpatient psychiatric care).

Conclusions: Research to better understand racial disparities in suicide risk among preadolescents can inform prevention efforts.

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Dr. Hammond serves on the scientific advisory board of Forbes & Manhattan. The other authors report no financial relationships with commercial interests.

A subset of these data was presented at the American Academy of Child and Adolescent Psychiatry Virtual Annual Meeting, October 12–24, 2020.

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The number of emergency department (ED) visits for suicidal thoughts and behaviors among American youths has increased dramatically over the past decade (1), particularly among preadolescents. This trend has called attention to national-level strains on ED services, which often have limited resources to effectively manage pediatric mental health emergencies (2–4). From 2001 to 2015, the suicide rate among Black children in the United States nearly doubled (5, 6), and rates of suicide were approximately twice as high among Black children compared with White children younger than age 13 (7, 8). Although definitive explanations remain elusive, potential reasons for racial disparities in suicide rate include higher exposure to adversity and discrimination, greater access to firearms, and reduced access to and utilization of mental health services among Black children compared with those of other racial groups (9–11). Previous research has identified racial differences in clinical presentation among children at risk of suicidal behaviors, with Black children being less likely to report suicidal ideation than White peers (12). This study examined racial differences in ED visit characteristics and management of suicidal thoughts and behaviors among preadolescents seen for a mental health crisis in an urban pediatric ED.

METHODS

We analyzed electronic medical record (EMR) data from patients ages 8–12 years visiting a quaternary care urban pediatric ED with a psychiatric-related chief complaint in 2019. This pediatric ED, located in the Mid-Atlantic region, has an estimated annual census of 35,000 patient visits (60% Black, 21% White, 10% Latino/Latina or Hispanic; approximately 60% had public insurance). An Epic EMR query identified all patients visiting the pediatric ED who had a psychiatric-related chief complaint during the study period. Variables of interest were extracted from the EMR and included demographic data, mode of arrival to the ED, urban residence (defined as a zip code within the city limits), primary ED clinical impression, and disposition from the ED. Additional extracted variables included item-level results from the Ask Suicide-Screening Questions (ASQ) tool (13), a four-item suicide risk screening instrument completed in the ED.

Three authors (C.V., L.M.R., and T.L.N.) classified the various clinical impression categories of the ED visit into the subcategories of suicidal thoughts or behaviors (i.e., having suicidal ideations or a recent suicide attempt), other externalizing disorders and behaviors (i.e., attention-deficit hyperactivity disorder, oppositional defiant disorder, aggressive behaviors, or disruptive behavior disorders and related problems or impairment), other internalizing disorders and symptoms (i.e., depression, anxiety, grief reactions, panic attacks, other mood disorder, or symptom-related problems or impairment), medical concerns, and other (i.e., substance use–related presentation, accidental ingestion, child abuse) as the primary reason for the ED visit.

Descriptive and comparative chi-square analyses by race were calculated with IBM SPSS Statistics, version 27.0. The analyses examined suicidal thoughts and behaviors as derived from the ASQ, primary ED visit clinical impression categories, and ED disposition in the total sample and by race (Black, White, and other or multiracial). Further descriptive and comparative analyses of demographic characteristics and ASQ responses by arrival mode

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and by diagnostic category were conducted. This study was approved by the Johns Hopkins University Institutional Review Board.

RESULTS

Our study sample of 504 preadolescents ages 8–12 years (mean±SD=10.6±1) represented 31% of the 1,646 patients with a psychiatric-related chief complaint who visited the pediatric ED in 2019. The preadolescents in this age group were predominantly male and Black, resided in an urban area, and arrived by private transportation. The most common ED clinical impression category noted was externalizing behaviors (28%, N=142), followed by suicidal thoughts and behaviors (24%, N=120) and other internalizing problems (10%, N=48), with the remainder of clinical impressions being categorized as medical concerns or other (nonspecified). Approximately 79% (N=398) completed the ASQ, with no racial differences in response rates. Of responders, 42% reported wishing they were dead (item 1), 32% reported feeling they would be better off dead (item 2), and 37% reported thinking of killing themselves in the past week (item 3) (Table 1). A previous suicide attempt (item 4) was reported by 28% of preadolescents. Of those who answered yes to any of the first four items of the ASQ, nearly one in five (19%) reported current thoughts of killing themselves (item 5, only administered after at least one response of yes to items 1–4). Most patients (89%, N=446) were discharged.

In comparative analyses, we found statistically significant racial differences in mode of arrival to the ED, ED clinical impression, suicidal thoughts and behaviors, and ED disposition. Black preadolescents were more likely to live in an urban area (80%, N=275 of 345) and arrive by police transportation (35%, N=122 of 345) than preadolescents in other racial groups. Clinical impressions for Black preadolescents were less likely to be categorized as suicidal thoughts and behaviors or other internalizing disorders and symptoms. Black preadolescents were also more likely to be discharged from the ED (93%, N=321 of 345). When examining only patients who responded yes to ASQ item 5, regarding being currently suicidal, we found that the differences by race persisted, with Black preadolescents being statistically significantly more likely to be discharged from the ED (82%, N=18 of 22) than their White (53%, N=8 of 15) and multi-racial (40%, N=2 of 5) peers and less likely to be admitted to inpatient psychiatric care or transferred to another facility (p=0.044) (Table S1 in the online supplement to this report).

Regarding suicidal thoughts and behaviors, Black preadolescents were less likely to report wishing they were dead (35%), feeling that they would be better off dead (27%), and thinking about killing themselves in the past week (33%) than were those of other racial groups, but their reported history of past suicide attempts did not differ from the histories of other racial groups (Table 1).

Further analyses of demographic characteristics and ASQ item comparisons by arrival mode (Table S2 in the online supplement) showed statistically significant differences in mode of arrival by gender (p<0.001), urban residency (p<0.001), ED clinical impression (p<0.001), wishes to be dead (p=0.003), and feelings of being better off dead (p=0.045). Boys were more likely than girls to be brought to the ED by police (63%, N=90 of 142 vs. 37%,

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N=52 of 142, respectively), whereas girls were more likely than boys to be brought to the ED by public and other transportation (67%, N=8 of 12 vs. 33%, N=4 of 12, respectively). Urban residents were more likely than nonurban residents to be brought to the ED by public transportation (92%, N=11 of 12 vs. 8%, N=1 of 12, respectively) and by police (90%, N=128 of 142 vs. 10%, N=14 of 142, respectively) and were less likely to be brought to the ED by private transportation compared with other forms of transportation (56%, N=164 of 293). Patients with externalizing symptoms were more likely to be brought to the ED by police (42%, N=60 of 142) or by ambulance (33%, N=19 of 57) compared with other forms of transportation. Patients with suicidal thoughts and behaviors were more likely to be brought to the ED by public transportation (33%, N=4 of 12), private transportation (29%, N=85 of 293), or ambulance (23%, N=13 of 57) than by police. Patients with other internalizing symptoms and behaviors were more likely to be brought to the ED by public transportation (25%, N=3 of 12) and private transportation (12%, N= 36 of 293) compared with other forms of transportation. Those brought to the ED for substance use—presumably accidental ingestion—or maltreatment, categorized as other presentations, were most often brought to the ED by private transportation (36%, N=104 of 293).

As might be expected, preadolescents who responded yes to the ASQ question about having current thoughts of killing themselves were most likely to be given an ED clinical impression of suicidal thoughts and behaviors (67%, N=28 of 42). Of those who had attempted suicide in the past, 15% (N=16 of 110) had a diagnosis of externalizing disorder (Table S3 in the online supplement).

DISCUSSION

More than one in four preadolescents visiting an urban ED with a behavioral health–related complaint reported a previous suicide attempt. Black preadolescents differed from non-Black peers in mode of arrival to the ED, reports of suicidal thoughts, ED clinical impression, and ED disposition. Consistent with national data (12), Black preadolescents were significantly less likely to report suicidal thoughts in the preceding week, but they reported rates of lifetime suicide attempts similar to those of other racial groups. Although individual factors such as impulsivity and related comorbid conditions (i.e., attention-deficit hyperactivity disorder) could account for this disparity (9, 10), cultural differences related to mental health stigma, mistrust of traditional mental health services, and neighborhood and structural exposures could also influence the experience or reporting of suicidal thoughts by Black preadolescents (14, 15).

In the urban ED in this study, non-Black preadolescents were three times as likely as Black children to be admitted to inpatient psychiatric care. Although racial differences in admission to inpatient psychiatric care could be related to the lower frequency of self-reported suicidal thoughts among Black preadolescents, other patient, parent, and provider factors likely contribute to this disparity, given that Black preadolescents in our sample who reported thoughts of suicide were still less likely to be hospitalized than were non-Black peers with thoughts of suicide.

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Black preadolescents were also more than twice as likely as preadolescents from other racial groups to have been transported to the ED by police or ambulance. Being male, being an urban resident, and having externalizing symptoms were also associated with being brought to the ED by police. These findings do not explain the reasons why police more frequently provided transportation to Black male preadolescents. These reasons could range from disparities in access to transportation to emergency medical services transport protocols in urban areas to criminalization of mental health–related behaviors among Black males. Additional research is warranted to explain these disparities.

Limitations of this study included its relatively small sample, its cross-sectional nature, and its focus on a single ED. In addition, these data did not allow for an exploration of differences between patients who responded to the ASQ and those who did not or for an examination of the reasons for the lower rate of ASQ administration to younger children. Future examination of the ED referral source (e.g., school setting vs. home) could also help explain the differences in mode of arrival. Finally, these data were from a period of only 1 year, and they were from patients visiting the ED before the COVID-19 pandemic.

ED providers are in a unique position to prevent suicidal behaviors in at-risk groups. Because most preadolescent patients evaluated for suicidal thoughts and behaviors in the ED were discharged, there is reason to develop brief behavioral ED interventions and effective linkages with community-based mental health services. Moreover, children who experience higher exposure to overall stress may more often seek services in the ED and benefit more from ED-based interventions. These findings highlight the need for research focusing on culturally tailored strategies to identify and support youths with suicidal thoughts and behaviors in ED settings.

Proposed culturally tailored interventions for Black children involve screening for depression in outpatient settings and the use of evidence-based interventions previously studied with Black youths (12). Although the body of research is scant, treatments such as multisystemic therapy to reduce suicidal thoughts and attempts and attachment-based family therapy for suicidal thoughts have proven to be effective for Black youths. Although they have not been sufficiently tested with Black youths, cognitive-behavioral therapy and interpersonal psychotherapy are established therapies for depression and suicide risk among adolescents, with less evidence of their efficacy with children. More research is needed to better understand whether best practices for suicide prevention, such as brief interventions linked to suicide risk screening, safety planning, motivational interviewing, and caring contacts are effective with Black youths.

CONCLUSIONS

Compared with other racial groups, Black preadolescents visiting a pediatric ED for mental health concerns were less likely to report suicidal thoughts, despite equivalent lifetime histories of past suicide attempts, and were more likely to be brought to the ED by police and to be discharged rather than admitted to inpatient psychiatric care. More research is needed to guide policies and programs for suicide prevention among Black preadolescents.

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Cultural adaptations of evidence-based practices to reduce suicide risk among Black children should be explored.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Acknowledgments

Dr. Vidal currently receives research grant funding from the American Academy of Child and Adolescent Psychiatry (Physician Scientist Career Development Award K12DA000357). Dr. Hammond receives grant support from the Johns Hopkins Consortium for School-based Health Solutions and the Johns Hopkins University School of Medicine.

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HIGHLIGHTS

- Black preadolescents visiting a pediatric emergency department (ED) for behavioral concerns were significantly less likely to report suicidal thoughts than those of other racial groups, despite equivalent lifetime histories of past suicide attempts.
- Black preadolescents were more likely than their same-age peers of other races to be brought to the ED by police and to be discharged instead of being admitted to inpatient psychiatric care.
- Racial differences in admission to inpatient psychiatric care could be related to the lower frequency of self-reported suicidal thoughts among Black preadolescents, but other patient, parent, and provider factors may have contributed to this disparity.

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TABLE 1.

Comparison of positive responses to Ask Suicide-Screening Questions (ASQ) items, by race[a]

	Total sample		Race						
			White		Black		Other or multiracial[b]		
ASQ item	**N**	**%**	**N**	**%**	**N**	**%**	**N**	**%**	χ^2[c]
1. In the past few weeks, have you wished you were dead?[d]	165	42	55	57	95	35	15	50	14**
2. In the past few weeks, have you felt that you or your family would be better off if you were dead?[e]	124	32	37	38	72	27	15	50	9*
3. In the past week, have you had thoughts about killing yourself?[f]	148	37	47	47	88	33	13	43	7*
4. Have you ever tried to kill yourself?[g]	110	28	26	27	75	28	9	29	1
5. Are you having thoughts of killing yourself right now?[h]	42	19	15	22	22	16	5	29	3

[a]Response options were yes or no, and the reference response to all items was no.

[b]"Other" includes Asian and Native American.

[c]For all, df=2.

[d]Total number of answers received: total, N=395; White, N=97; Black, N=268; other or multiracial, N=30.

[e]Total number of answers received: total, N=392; White, N=97; Black, N=265; other or multiracial, N=30.

[f]Total number of answers received: total, N=400; White, N=100; Black, N=270; other or multiracial, N=30.

[g]Total number of answers received: total, N=396; White, N=96; Black, N=269; other or multiracial, N=31.

[h]ASQ item 5 (also known as the ASQ follow-up item) was administered to all respondents who screened positive by responding yes to any of the ASQ screening items (items 1–4). Total number of answers received: total, N=221; White, N=67; Black, N=137; other or multiracial, N=17.

*$p \leq .05$,

**$p \leq .001$.